

KINESIS

A Fraud-Resistant, Self-Sustaining Play-to-Earn & Move-to-Earn Web3 Ecosystem

Abstract (Short Form)

Kinesis is a play-to-earn and move-to-earn decentralized application (dApp) that lets users log in with a crypto wallet, play skill-based games, stake tokens, and earn rewards for real-world walking tracked via their phone. Platforms in this space (Sweatcoin, StepN) have proven demand but consistently fail on token sustainability and reward fraud. Kinesis is engineered specifically to solve both, while also solving onboarding friction and platform revenue transparency — problems largely ignored by comparable systems.

Where Current Apps Fail

- Unsustainable tokenomics — StepN's token value collapsed over 90% once reward payouts outpaced real demand.
- Reward fraud — bots and GPS-spoofers fake activity to farm rewards, undetected.
- Poor onboarding — users must already own/understand crypto gas fees before using the app.
- No transparent revenue model — unclear how the platform itself sustains financially.

Problem → Our Solution

- **Token collapse** → Dual-token model (scarce governance token + inflationary utility token) with engineered spending sinks to balance earning and spending.
- **Fraud** → AI model fuses phone sensor plausibility (accelerometer/GPS) with on-chain wallet behaviour to catch fakes before rewarding.
- **Onboarding friction** → Account abstraction (ERC-4337): wallet created silently behind a normal login; gas fees sponsored via a paymaster.
- **Unclear revenue** → Transparent on-chain treasury contract collecting fees, royalties, and sponsorships, visible live to anyone.
- **Unverifiable records** → Every action logged immutably on a Hyperledger Fabric audit ledger.
- **Rigged outcomes** → Chainlink VRF generates game randomness off-chain with on-chain cryptographic proof — provably fair.
- **Centralized control** → DAO voting lets token holders approve reward-rate/game changes, preventing unilateral rug-pulls.

Core Features

- Wallet-based login (no passwords) + skill-based stake-to-play games (trivia/puzzle/prediction).
- Staking module for passive token yield.
- Mobile move-to-earn app using built-in phone sensors (no extra IoT hardware needed).
- AI-driven anti-cheat / behavioural fingerprinting (physical + financial signal fusion).
- Dual-token economy with engineered sinks and faucets.
- Hyperledger Fabric dual-ledger audit layer, alongside a public test-network payment chain.
- Gasless onboarding via ERC-4337 account abstraction.
- DAO-based governance for reward rates and new games.
- Provably-fair randomness via on-chain oracle (Chainlink VRF).
- Reward redemption for Amazon gift vouchers alongside crypto.

How the Platform Earns

- Small transparent fee (rake) on every game and staking pool.
- Royalty on secondary sales of in-app cosmetic NFTs.
- Brand/sponsorship partnerships for reward pools and vouchers.
- All revenue streams accumulate publicly in an on-chain treasury contract.

Technology Stack

Solidity smart contracts (Polygon Amoy testnet) • Hyperledger Fabric • Python/Flask backend • React/Next.js web dApp • React Native/Flutter mobile app • Chainlink VRF • Docker • AI/ML behavioural fingerprinting model.